

AnchorHOI: Anchoring Monocular Hand-Object Reconstruction with Discrete and Spatial Priors

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Abstract. Monocular hand-object interaction (HOI) reconstruction is essential for understanding human manipulation. However, existing methods typically optimize object pose and shape in a largely unconstrained continuous space, making them highly susceptible to local minima and catastrophic tracking failures under in-hand occlusions and depth ambiguities. In this paper, we present AnchorHOI, a novel prior-guided optimization framework that structurally tames this ill-posed continuous space. Our methodology features three core innovations: First, we bypass the fragility of continuous 6-DoF pose regression by reformulating in-hand object pose estimation as a finite template-based search problem. By retrieving and geometrically verifying discrete pose hypotheses using foundation model features, we convert brittle continuous optimization into a robust search-and-refine paradigm. Second, built upon these stabilized poses, we deploy an efficient dynamic HOI modeling approach based on fast neural implicit surfaces. Combined with an occlusion-aware supervision strategy, this enables coherent multi-view surface aggregation across long sequences, achieving a 112 $\times$ speedup over recent dynamic reconstruction methods. Finally, to resolve monocular depth ambiguity, we explicitly anchor the global hand-object alignment using a 3D relative offset prior. Extensive experiments demonstrate that AnchorHOI sets a new state-of-the-art in HOI reconstruction stability, efficiency, and physical plausibility. Code will be released.

Keywords: 3D Reconstruction · Hand–Object Interaction · Neural Rendering

1 Introduction

Reconstructing hand–object interactions (HOI) involves recovering the 3D geometry and pose of both hands and manipulated objects from visual observations such as images or videos. This problem is fundamental to understanding human manipulation and interaction with the physical world, and supports a wide range of applications including in-hand object scanning [24], robotic manipulation [53], and AR/VR [1].

Recent works typically formulate HOI reconstruction as optimization-based pipelines [15, 23, 24, 37, 60] in part due to the scarcity of HOI datasets. These

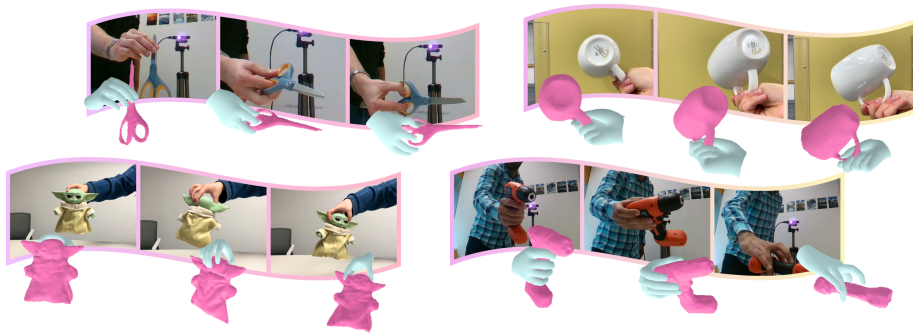


Fig. 1: Given monocular video input, AnchorHOI reconstructs hand-object interactions via a prior-guided optimization pipeline. By constraining the solution space with discrete pose hypotheses and spatial priors, our method enables stable object pose estimation and coherent 3D reconstruction under in-hand occlusions.

pipelines generally involve several stages, such as estimating object pose, reconstructing object geometry, and aligning the hand and object to recover the interaction. However, achieving stable and coherent HOI reconstruction remains challenging. In many existing pipelines, object pose and global hand-object alignment are treated as unconstrained continuous variables, which often leads to unstable optimization under monocular observation and frequent occlusion: (1) **Unstable Object Pose Estimation.** Frequent hand-object occlusions and diverse category-agnostic object geometries make recovering object poses difficult. Huang et al. [23] assume no relative shift between the hand and the object, and therefore infer object pose directly from hand-based priors [48]. However, this assumption restricts the modeling of dynamic manipulation. Other methods such as HOLD [15] and MagicHOI [60] rely on Structure-from-Motion (SfM) initialization. However, SfM often degrades under in-hand occlusion or textureless objects. (2) **Ambiguous Global Hand-Object Alignment.** Inherent monocular depth ambiguity further complicates consistent hand-object placement. Prior works [15, 37] mainly supervise hand-object alignment through 2D mask consistency. However, 2D consistency does not guarantee geometrically valid placement in 3D space. As a result, the hand and object may appear aligned in 2D while remaining spatially separated in 3D due to depth-scale ambiguity.

In this paper, we propose **AnchorHOI**, a prior-guided optimization method for monocular HOI reconstruction. Our key idea is that the feasible solution space should first be structured using predictive geometric priors, rather than directly optimizing in an unconstrained continuous space. Unlike prior approaches that treat feed-forward priors primarily as regularization losses, we use them as direct constraints that restrict the search space to a compact set of physically and geometrically plausible hypotheses, within which optimization refines the final solution. We introduce this idea at two critical stages: (1) **Finite Template Pose Hypothesis Search.** Instead of directly regressing object pose in the continuous $SE(3)$ space, we reformulate pose estimation as a finite template search problem. A feed-forward geometric prior defines a discrete set of can-

didate poses, effectively discretizing the solution space and providing a robust initialization that stabilizes pose recovery under in-hand occlusion. **(2) Hand-Object Alignment via 3D Relative Prior.** To resolve monocular depth ambiguity, we replace unconstrained global alignment with a 3D relative offset constraint. With the relative spatial relationship between the hand and object, the alignment search space is restricted to a physically plausible region, enabling spatially consistent hand-object interactions beyond what 2D supervision alone can guarantee.

Specifically, we instantiate this idea in a three-stage pipeline. First, we perform constrained object pose estimation via a finite template hypothesis search, where candidate poses are selected from a prior-defined discrete template set and refined locally, converting continuous $SE(3)$ regression into a structured search-refine procedure. Second, with stabilized object poses, we perform efficient implicit object reconstruction in a canonical object coordinate frame, aggregating multi-frame observations for stable multi-view optimization. This design enables efficient reconstruction across long HOI sequences and achieves a $112\times$ speedup over recent HOI reconstruction methods [15]. Finally, we resolve global hand-object alignment ambiguity using a relative offset prior that constrains hand translation in object space. Within this constrained alignment space, we refine hand parameters under physically motivated interaction constraints to ensure spatially consistent and physically plausible hand-object interactions.

In summary, our main contributions are:

- We propose **AnchorHOI**, a prior-guided optimization framework for monocular HOI reconstruction that restructures the feasible solution space using predictive priors, enabling stable and efficient optimization under hand-object occlusion and monocular ambiguity.
- We introduce a **finite template hypothesis search** for object pose estimation, which discretizes the pose space and stabilizes pose recovery under hand-object occlusion.
- We develop an **efficient reconstruction framework** that aggregates multi-frame observations for stable object reconstruction in long HOI sequences, achieving up to $112\times$ speedup over recent HOI reconstruction methods.
- We introduce a **3D relative offset-guided hand-object alignment strategy** that constrains global hand placement in object space, resolving monocular depth ambiguity and enabling spatially consistent and physically plausible interaction reconstruction.

2 Related Works

Object reconstruction and generation. Recent advances in 3D reconstruction from 2D images have made remarkable progress [5, 6, 29, 39, 57, 59, 64, 67]. Classical methods rely on multi-view stereo [16, 27], while recent works employ neural representations for more accurate reconstructions. However, most still depend on SfM [4], which are not stable under dynamic, low-textured scenes [23].

Data-driven approaches leverage large-scale datasets and generative priors to infer 3D structure directly from images, such as feed-forward reconstruction models and diffusion-based 3D generation frameworks [9, 14, 22, 28, 30, 31, 34–36, 38, 44, 52, 55]. Our approach leverages advantages from both neural representations [61] and feed-forward models [63] to tackle the challenging task of reconstructing not only objects but also hands from monocular videos.

Hand-held object reconstruction. Reconstructing hand-held objects has gained increasing attention in recent years, yet remains highly challenging due to frequent hand occlusions and the wide variety of unseen object categories. Many approaches assume that the object shape is known and focus on estimating its pose [4, 65]. While effective, ground-truth object templates are typically unavailable in real-world settings. Learning-based methods [21, 26] attempt to predict both object shape and poses directly, but they generalize poorly to unseen object shapes. Consequently, recent works explore neural representations to improve object shape generalization. Huang et al. [23] leverage the MANO hand prior [48] to approximate object poses in static settings. BundleSDF [62] and Hampali et al. [18] extend reconstruction to dynamic interactions by jointly optimizing the object mesh and its pose trajectory. HOLD [15] and MagicHOI [60] further improve dynamic reconstruction quality, but rely on SfM initialization, which is unstable under severe occlusions. Chen et al. [7] track an implicit object shape by aligning per-frame point clouds to its SDF surface, but require access to the object point cloud at every frame. Most related to our work, Jiang et al. [24] leverage a text-to-3D generative model to synthesize a pseudo object template, reducing the difficulty of reconstructing unknown objects. However, optimizing the pseudo-template pose in continuous 6-DoF space is inherently unstable and often results in degraded reconstructions. We instead formulate this step as a discrete template-matching problem, followed by a refinement stage for accurate pose estimation in continuous space.

Hand-object interaction reconstruction. Beyond accurate object shapes and poses, hand-object interaction reconstruction additionally aims to capture the geometric relationship between object and hands in the 3D space [4, 11, 13, 40, 56, 65, 69]. Most existing pipelines build upon 3D hand estimation [8, 10, 25, 32, 33, 43, 45, 46, 49, 66, 70–73] as an intermediate representation. While prior works suffer from inconsistent initialization of hand and object coordinate systems, methods such as HOLD [15] and EasyHOI [37] tackle this issue using 2D mask supervision for global alignment followed by refinement. However, mask-based alignment enforces only silhouette overlap, often producing hand-object pairs that appear aligned in the 2D camera view but remain misaligned in 3D space. We observe that the relative hand-object relationship remains invariant across coordinate systems and can therefore serve as a geometric prior. We leverage a feed-forward method [12] to estimate this relationship, aligning the hand parameters within the object coordinate frame for consistent interaction reconstruction.

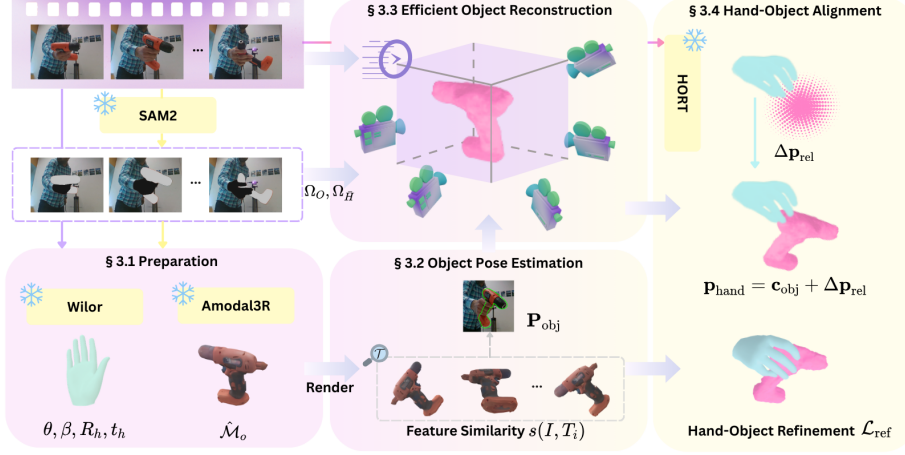


Fig. 2: Overview of AnchorHOI. Given a monocular HOI video, we initialize the hand parameters and generate a proxy object mesh (§3.1). Object poses are estimated via a finite template search with feature retrieval and geometric verification (§3.2), enabling stable initialization under in-hand occlusion. Using these poses, we perform efficient implicit object reconstruction in a canonical coordinate frame with occlusion-aware supervision (§3.3). Finally, a 3D relative offset prior anchors the hand relative to the object for physically consistent interaction refinement (§3.4).

3 Method

Monocular HOI reconstruction is severely underconstrained due to frequent occlusion and depth ambiguity between the hand and object. Instead of directly optimizing in the full continuous space, we systematically introduce feed-forward geometric priors to constrain the nonlinear optimization space at key stages. Specifically, we introduce constraints into (1) object pose estimation via finite template search to stabilize pose under occlusion (§3.2), and (2) hand-object alignment via a 3D relative offset prior to resolve spatial ambiguity (§3.4). Following pose stabilization, we perform efficient implicit object reconstruction (§3.3), which leverages the stabilized object poses for multi-view optimization and establishes a canonical object frame for subsequent alignment. Overall, our design converts the original unconstrained optimization formulation into a stable and computationally efficient reconstruction pipeline.

3.1 Preparation

Object Prior for Finite Template Search. Given monocular HOI videos as input, the manipulated object is typically unknown and no CAD model is available. We therefore generate an initial object mesh $\hat{\mathcal{M}}_o$ using Amodal3R [63], which serves as a geometric proxy for rendering pose templates used in the finite template search.

Hand Initialization. We initialize the hand parameters [48] using the off-the-shelf model Wilor [45], including hand pose $\theta \in \mathbb{R}^{45}$, hand shape $\beta \in \mathbb{R}^{10}$, global rotation $R_h \in SO(3)$, and translation $\mathbf{t}_h \in \mathbb{R}^3$.

3.2 Pose Estimation via Finite Template Hypothesis Search

Given the proxy mesh $\hat{\mathcal{M}}_o$, we reformulate object pose estimation as a finite pose template selection problem (Fig. 3). Instead of directly optimizing the object pose $\mathbf{p}_{obj} \in SE(3)$ in continuous space, we discretize the pose space by defining a discrete pose set $\mathcal{P} = \{\mathbf{p}_i^{\text{temp}}\}_{i=1}^m$, where m denotes the number of discrete pose hypotheses. Using $\hat{\mathcal{M}}_o$, we render a corresponding template set $\mathcal{T} = \{T_i\}_{i=1}^m$, where each template T_i is associated with pose $\mathbf{p}_i^{\text{temp}}$. For each input video frame I , we first select the best-matching template T_{win} from $\mathcal{T}$ and use its associated pose $\mathbf{p}_{\text{win}}^{\text{temp}}$ as a coarse pose hypothesis. Starting from this discrete initialization, we further perform continuous pose refinement to bridge the discretization gap and obtain the final pose estimate. This formulation restricts the optimization to a small set of plausible pose hypotheses and improves robustness under hand-object occlusions.

Feature Representation for Template Matching. To bridge the domain gap between rendered templates and real input images, we follow FoundPose [42] in using DINOv2 patch descriptors as the feature representation. Specifically, patch-level descriptors from DINOv2 are extracted for each rendered template $T_i \in \mathcal{T}$ and clustered via k -means to obtain a compact embedding. Given an input frame I , we compute cosine similarity between the embeddings of the input frame and templates:

$$s(I, T_i) = \frac{\langle f(I), f(T_i) \rangle}{\|f(I)\| \|f(T_i)\|}, \quad (1)$$

where $f(\cdot)$ denotes the clustered DINOv2 embedding.

Feature-Guided Pose Search with Geometric Verification. Since our input consists only of monocular HOI videos, the object geometry is approximated using a generated proxy mesh $\hat{\mathcal{M}}_o$ rather than an accurate CAD model. Because the proxy mesh may differ from the real object in both shape and appearance, in practice we observe that the template with the highest feature similarity does not necessarily correspond to the most geometrically consistent pose. We therefore adopt a feature-guided discrete pose search strategy with geometric verification.

First, we use DINOv2 feature similarity to retrieve a small candidate set $\mathcal{T}_{\text{cand}}(I)$ from the template set:

$$\mathcal{T}_{\text{cand}}(I) = \text{Top-}n(\{s(I, T_i)\}_{i=1}^m). \quad (2)$$

Each candidate template $T_i \in \mathcal{T}_{\text{cand}}(I)$ is associated with a known pose $\mathbf{p}_i^{\text{temp}}$, which we treat as a discrete pose hypothesis. We then perform geometric verification by running RANSAC+PnP on 2D-3D correspondences established via DINOv2 patch matching, where 2D points are sampled on a dense grid inside the object mask and 3D points are the corresponding template-associated object

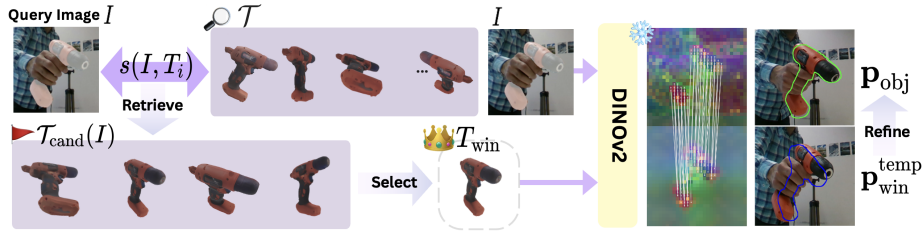


Fig. 3: Object Pose Estimation. Given a query image I , we **1) Retrieve** the top- n candidate templates $\mathcal{T}_{\text{cand}}(I) \subset \mathcal{T}$ with predefined poses from the template set $\mathcal{T}$ using feature similarity $s(I, T_i)$. We then **2) Select** the geometrically consistent template T_{win} via geometric verification. The associated template pose $\mathbf{p}_{\text{win}}^{\text{temp}}$ provides a initialization, which is further **3) Refined** by minimizing reprojection error to obtain the final object pose $\mathbf{p}_{\text{obj}}$.

coordinates, and select the hypothesis that maximizes the inlier count under a reprojection threshold. This strategy keeps the correct pose within a small candidate set even when the most similar template is not geometrically optimal. Unlike FoundPose, which directly estimates object pose from patch-level 2D–3D correspondences, we instead treat template poses as discrete pose hypotheses from a finite set and use RANSAC+PnP only for geometric verification to select among the retrieved candidates. This formulation improves robustness when the proxy mesh is imperfect or when hand–object occlusions break reliable correspondences.

Continuous Pose Refinement. The pose of the best-ranked template $\mathbf{p}_{\text{win}}^{\text{temp}}$ provides a coarse initialization of the object pose. Starting from this template, we perform continuous pose refinement by minimizing reprojection error between the rendered template T_{win} and the observed image features. This refinement step allows the discrete template pose to converge to a precise continuous estimate, yielding the final object pose $\mathbf{p}_{\text{obj}}$. All poses are estimated independently for each frame without temporal smoothing.

3.3 Efficient Reconstruction with Occlusion-Aware Supervision

Given the estimated object poses, we reconstruct object geometry and appearance using an implicit signed distance field (SDF) representation. Following object-centric neural surface reconstruction [15, 24], all frames are transformed into a canonical object coordinate system using the estimated object poses, enabling multi-view observations to be aggregated in a unified space.

Although recent neural surface formulations enable efficient reconstruction in static settings [61], applying them to dynamic HOI sequences remains challenging (see *Supplementary Material* for further discussion). Object observations are frequently occluded by the hand during manipulation, resulting in incomplete and unreliable supervision. To address this issue, we introduce an occlusion-aware supervision strategy that explicitly controls where appearance

and semantic constraints are applied during optimization. For each frame, we obtain segmentation masks using SAM2 [47]. Let p denote a pixel and m_p its predicted semantic label (background, object, hand). We define the object region $\Omega_O = \{p : m_p = \text{object}\}$ and the non-hand region $\Omega_H = \{p : m_p \neq \text{hand}\}$.

During hand–object interaction, hand masks may partially overlap with object regions due to occlusion. Applying semantic supervision in these regions may introduce incorrect supervision and destabilize geometry optimization. We therefore apply semantic supervision only on non-hand pixels:

$$\mathcal{L}_{\text{sem}} = \frac{1}{|\Omega_H|} \sum_{p \in \Omega_H} \|\hat{y}(p) - y(p)\|_2^2,$$

where $\hat{y}(p) \in \mathbb{R}^K$ denotes the predicted semantic output and $y(p) \in \{0, 1\}^K$ is the one-hot label from the SAM2 mask. Excluding hand regions avoids unreliable supervision in occluded areas and improves optimization stability.

To refine object appearance while avoiding contamination from hand or background pixels, RGB supervision is applied only on object pixels:

$$\mathcal{L}_{\text{rgb}} = \frac{1}{|\Omega_O|} \sum_{p \in \Omega_O} \|\hat{c}(p) - c(p)\|_1,$$

where $\hat{c}(p) \in \mathbb{R}^3$ and $c(p) \in \mathbb{R}^3$ denote the predicted and observed RGB values.

The overall reconstruction objective is:

$$\mathcal{L} = \mathcal{L}_{\text{rgb}} + w_{\text{sem}}(t) \mathcal{L}_{\text{sem}} + \lambda_{\text{eik}} \mathcal{L}_{\text{eik}} + w_{\text{sparse}}(t) \mathcal{L}_{\text{sparse}},$$

where $\mathcal{L}_{\text{eik}}$ enforces the Eikonal constraint [59] and $\mathcal{L}_{\text{sparse}}$ regularizes density sparsity [17]. The weights $w_{\text{sem}}(t)$ and $w_{\text{sparse}}(t)$ are scheduled during training.

Notably, our design achieves state-of-the-art reconstruction quality while enabling efficient and stable reconstruction for long HOI sequences. For each video sequence, our reconstruction completes in approximately 16 minutes, achieving a **112×** speedup over the state-of-the-art HOI reconstruction method [15].

3.4 Hand-Object Alignment via 3D Relative Prior

This stage ensures spatially consistent and physically plausible hand–object interaction. A learned 3D relative prior stabilizes global hand placement, while physically motivated constraints refine contact and prevent interpenetration.

3D Relative Offset Prior. Since the hand and object are estimated independently under monocular input, their global placement in 3D space is ambiguous. Constraints based only on 2D projections cannot resolve this ambiguity and often lead to unstable hand–object alignment.

To provide explicit 3D information, we introduce a 3D relative offset prior between the hand and the object. Using HORT [12], we obtain the hand mesh and the object point cloud, from which we compute the hand palm position $\tilde{\mathbf{p}}_{\text{hand}}$ and the object center $\tilde{\mathbf{p}}_{\text{obj}}$ in a shared coordinate frame. Given the reconstructed

object center $\mathbf{c}_{\text{obj}}$ in the canonical object frame, we compute the hand translation as

$$\mathbf{p}_{\text{hand}} = \mathbf{c}_{\text{obj}} + (\tilde{\mathbf{p}}_{\text{hand}} - \tilde{\mathbf{p}}_{\text{obj}}) = \mathbf{c}_{\text{obj}} + \Delta\mathbf{p}_{\text{rel}}, \quad (3)$$

where $\Delta\mathbf{p}_{\text{rel}}$ denotes the predicted 3D hand-object offset. This prior anchors the hand position relative to the object before nonlinear refinement, preventing depth drift during optimization.

Hand-Object Interaction Refinement. After global alignment is constrained by the relative offset prior, we refine hand parameters within this stabilized space. We optimize the hand translation $\mathbf{t}_h$ and shape parameters β under physically motivated constraints:

$$\mathcal{L}_{\text{ref}} = \lambda_{\text{mask}}\mathcal{L}_{\text{mask}} + \lambda_{\text{contact}}\mathcal{L}_{\text{contact}} + \lambda_{\text{pene}}\mathcal{L}_{\text{pene}}, \quad (4)$$

where $\mathcal{L}_{\text{mask}}$ enforces silhouette consistency with off-the-shelf segmentations [47], $\mathcal{L}_{\text{contact}}$ encourages proximity between hand contact regions and the object surface following [15], and $\mathcal{L}_{\text{pene}}$ penalizes hand-object interpenetration. Operating within the offset-constrained alignment space ensures that these losses refine interaction details without reintroducing global depth ambiguity. See more details in the *Supplementary Material*.

4 Experiments

4.1 Implementation Details

We use SAM2 [47] to obtain both object and hand masks, providing clean and reliable segmentation cues for pose estimation and HOI optimization. For object pose estimation, we extract DINOv2 [41] ViT-g/14 features from the 30th layer. We empirically find that this intermediate layer offers an effective balance between local geometric detail and global semantic context, which is crucial for robust template matching under severe occlusions and low-texture conditions. During training, it is worth noting that we extend the efficient computation framework [61] to the dynamic HOI reconstruction pipeline, which enables processing a sequence of 400 frames on a single RTX 4090 GPU in approximately 16 minutes. More details are provided in the *Supplementary Material*.

4.2 Metrics and Datasets

Metrics. We evaluate our method across object reconstruction quality, hand pose accuracy, and hand-object interaction following the metrics adopted in prior works [15, 60, 68]. For object reconstruction, we apply Iterative Closest Point (ICP) alignment [2] to register the reconstructed mesh with the ground truth, and then compute the Chamfer Distance (CD) and F-score to evaluate geometric fidelity. CD measures point-to-point mesh discrepancy, while F-score (at 5 mm and 10 mm) reflects local shape accuracy after alignment. For hand pose

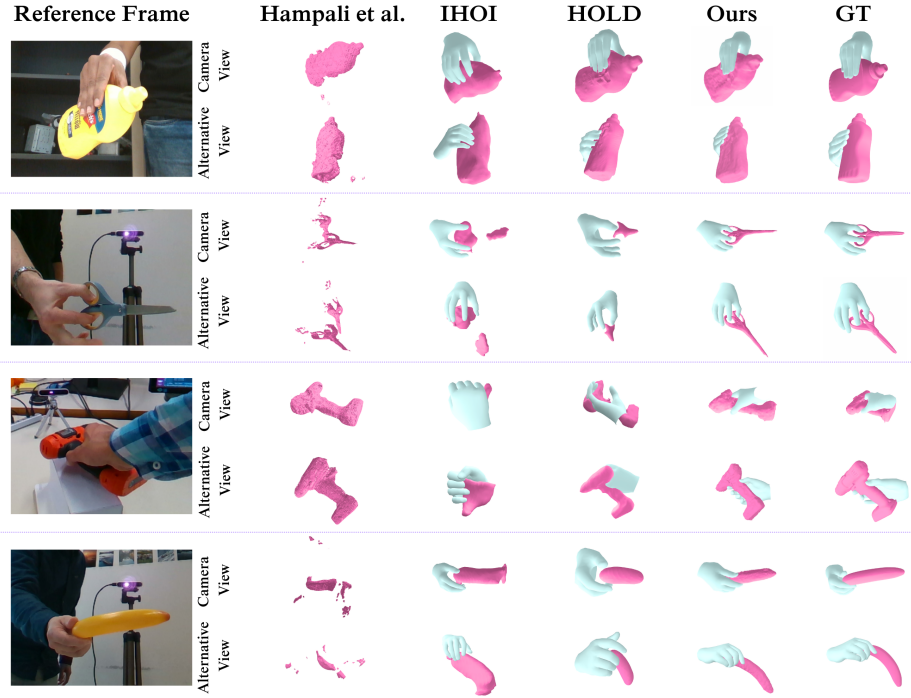


Fig. 4: Qualitative comparison with state-of-the-art methods. Compared with prior approaches, our method produces more stable and accurate hand-object reconstructions. Existing methods often fail on textureless objects (banana), thin structures (scissors), and heavily occluded interactions (power drill, mustard bottle), leading to pose instability, distorted geometry, or hand-object misalignment. In contrast, our method maintains consistent object poses and physically plausible hand-object interactions across these challenging scenarios.

accuracy, we use the Root-relative Mean-Per-Joint Position Error (MPJPE), measuring the mean joint distance after normalizing to the hand root. Following prior work [60], the hand pose parameters are initialized using an off-the-shelf hand pose estimator and remain fixed during the optimization. We report MPJPE for completeness and to maintain comparability with other HOI reconstruction works. Additionally, we report the hand-relative Chamfer Distance (CD_h) to measure the object’s shape and pose consistency relative to the hand. For ablation studies on the pose estimation, we additionally report the Absolute Trajectory Error (ATE) [54] to evaluate trajectory-level pose stability. Following standard practice in SLAM evaluation, we align the predicted trajectory with the ground-truth trajectory using Umeyama alignment [58], and compute the ATE in millimeters.

Datasets. We use the HO3D-v3 dataset [19, 20] for evaluation, which contains RGB videos of hand-object interactions along with ground-truth object anno-

tations from the YCB dataset [3]. Following the experimental setup of HOLD, we use the same set of sequences and preprocessing pipeline. To further assess real-world performance and generalization, we additionally capture several in-the-wild monocular videos with diverse objects and hand motions.

Table 1: Comparison with state-of-the-art HOI reconstruction methods. We evaluate object reconstruction accuracy, hand pose accuracy, and hand-object interaction quality. Our method achieves consistently superior performance, demonstrating significant improvements over prior approaches.

Method	CD [cm ²] ↓	F10 [%] ↑	MPJPE [mm] ↓	CD _h [cm ²] ↓
iHOI [68]	3.8	75.8	38.4	41.7
DiffHOI [69]	4.3	68.8	32.3	43.8
HOLD [15]	1.3	90.6	41.4	21.4
Ours	0.6	93.4	4.2	16.4

4.3 State-of-the-Art Comparison

Qualitative Comparison. Fig. 4 provides a qualitative comparison between our method and state-of-the-art approaches. Hampali et al. [18] represent the current in-hand object scanning paradigm, which focuses solely on reconstructing the object without modeling the hand. As the code is unavailable and only partial reconstruction sequences are publicly accessible, we follow their released sequences for comparison. IHOI [68] is a single-image method that reconstructs generic hand-object configurations without requiring category-specific 3D templates. HOLD [15] is the state-of-the-art video-based baseline that jointly reconstructs both the hand and object, and is therefore the most relevant method to ours.

Our qualitative comparisons show that existing methods consistently struggle with some challenging cases, including textureless objects (e.g., the banana) and thin objects (e.g., the scissors). In contrast, our method produces significantly more accurate and stable reconstructions. This improvement stems from our robust in-hand object pose estimation, which is less affected by missing textures, color ambiguities, or partial occlusions. As a result, our approach maintains high reconstruction fidelity even under these challenging conditions, which is one of the key distinctions between our method and previous baselines. Moreover, for everyday manipulation scenarios involving frequent hand-object contact or occlusions, such as the power drill and mustard bottle, our method also delivers superior reconstruction quality. By incorporating both spatial and physical consistency during our hand-object interaction optimization (§ 3.4), our results exhibit realistic interaction behaviors, effectively avoiding penetrations while preserving meaningful contact throughout the entire manipulation sequence.

Table 2: Comparison with state-of-the-art in-hand object scanning methods. All methods are evaluated on the same HO3D sequences [20] follow Hampali et al. [18] for fair comparison.

Method	CD [cm ²] ↓	F5 [%] ↑	F10 [%] ↑
Hampali et al. [18]	1.4	57.4	79.9
Jiang et al. [24]	0.6	76.0	94.4
Ours	0.5	78.2	94.4

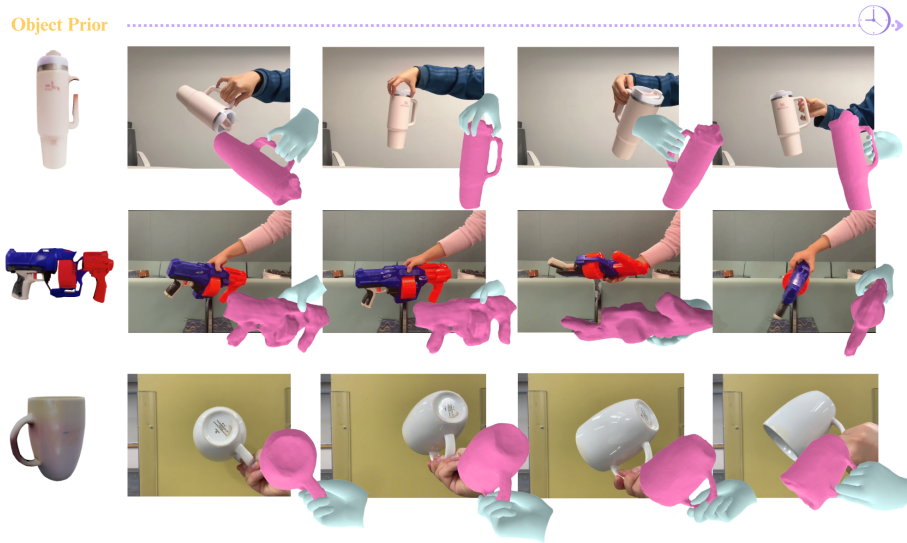


Fig. 5: In-the-wild qualitative results. We show the generated object prior used for pose estimation and the reconstructed results across temporally ordered frames of monocular videos. Our method produces stable poses and consistent reconstructions under diverse object geometries and real-world conditions.

Quantitative Comparison. We conduct quantitative comparisons against both hand-object reconstruction methods and in-hand object scanning methods, as summarized in Tab. 1 and Tab. 2. Tab. 1 reports object reconstruction quality, hand pose accuracy, and hand-object interaction accuracy. Across all metrics, our method achieves the best overall performance. Tab. 2 compares our method with current in-hand object scanning methods, focusing on object reconstruction quality. We evaluate on the video sequences used by Hampali et al. [18], and the results demonstrate that our method also performs favorably in the in-hand object scanning setting.

Table 3: Ablation study of object pose estimation. We evaluate the contribution of each key component in our pose estimation pipeline. Top-n retrieval, the template pose prior, and refinement all play essential roles in achieving stable and accurate in-hand object pose estimation.

Method	Top-n	Template	Refinement	CD [cm ²] ↓	F5 [%] ↑	F10 [%] ↑	ATE [mm] ↓
SfM Init	✗	✗	✗	5.8	60.4	75.2	4.1
w/o Top-n (Top-1 only)	✗	✓	✓	0.7	74.2	92.1	3.2
w/o Template Pose	✓	✗	✓	1.0	71.3	89.6	3.5
w/o Refinement	✓	✓	✗	1.1	71.8	87.8	3.5
Top-3 candidates	✓	✓	✓	0.6	76.2	92.5	3.0
Top-10 candidates	✓	✓	✓	0.7	76.0	91.9	3.0
Ours (full, Top-5)	✓	✓	✓	0.6	76.4	93.4	3.0

4.4 In-the-wild Results

We additionally capture several in-the-wild monocular videos to further assess the robustness of our method. As shown in Fig. 5, we present three representative cases: (1) a texture-less pink bottle with a slender geometry, (2) a toy water gun with rich geometric structures, and (3) a white mug that is largely texture-less and reflects colored environmental lighting. These examples demonstrate that our method can handle diverse object geometries and material properties, while remaining robust to imperfect generated object priors and challenging real-world conditions. Despite these factors, the method produces stable object poses and consistent reconstructions over time.

4.5 Ablation Study

Tab. 3 reports the ablation results for our object pose estimation pipeline, isolating the contribution of each key component. All variants are evaluated using the metrics defined in § 4.2, including CD, F5, F10, and ATE.

SfM Initialization. To evaluate the necessity of our pose estimation strategy via finite template search, we replace our pose estimation with a traditional SfM pipeline [50, 51], while keeping all subsequent reconstruction and optimization modules unchanged. SfM-based initialization leads to degraded CD, F5, F10, and ATE metrics, with the drop particularly evident on sequences with weak texture, repetitive patterns, or large hand-induced occlusions (see also Fig. 6). Since SfM relies on stable feature correspondences and long-range tracks, it struggles in HOI videos where occlusions and rapid object motion frequently break feature continuity, resulting in drifting or inconsistent object poses. In contrast, our template-based initialization remains stable under these conditions.

w/o Top-n (Top-1 Only). We disable Top-n template retrieval and directly select the template with the highest similarity scores, followed by pose refinement. Although DINOv2 features are robust, cosine similarity does not necessarily reflect geometric consistency, causing the top-1 template to often correspond to

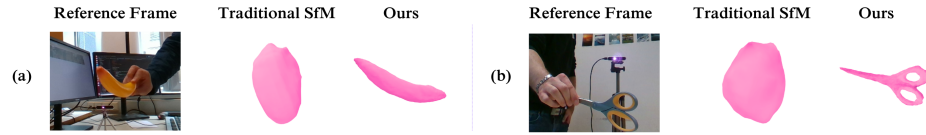


Fig. 6: Comparison between SfM initialization and our pose estimation method. SfM leads to unstable object poses under textureless or occluded conditions, while our method produces more consistent estimates and improved reconstructions.

an incorrect pose. The subsequent refinement step can alleviate small deviations but cannot correct a fundamentally wrong template choice, highlighting the importance of the Top- n candidate set and geometric verification.

w/o Template Pose. We disable the discrete template pose and estimate the pose solely using patch correspondences via PnP optimization. While correspondences and PnP can still recover a reasonable pose, the optimization becomes more vulnerable to local mismatches and partial occlusions without the template pose prior. This results in degraded accuracy, demonstrating that the template pose provides an essential coarse geometric prior rather than merely supplying correspondences.

w/o Refinement. We remove the continuous pose refinement and directly use the discrete template pose as the output. Since template poses belong to a discrete pose space and cannot perfectly match the input image, removing refinement retains the discretization error, causing a noticeable drop across all metrics. This highlights the importance of continuous refinement for accurate pose estimation.

Top- n Candidate Set Size. We evaluate Top-3, Top-5, and Top-10 and observe that all configurations outperform Top-1 by a clear margin. This confirms that, compared to selecting only the Top-1 template, constructing a candidate set and applying geometric consistency filtering consistently improves performance. Among these settings, Top-5 achieves the best overall accuracy, and we therefore adopt Top-5 as the default configuration in our full pipeline.

Summary. Across all settings, our ablations validate the importance of each module in our object pose estimation pipeline. Top- n retrieval and geometric consistency ensure reliable template selection, the template pose provides a strong coarse prior, and refinement is critical for precise optimization. Together, these components enable robust pose estimation even under occlusion and low-texture conditions.

5 Conclusion

In this paper, we introduce **AnchorHOI**, a prior-guided optimization framework for monocular hand-object interaction reconstruction. By constraining the nonlinear solution space with feed-forward priors, our method addresses the fundamental instability of monocular HOI reconstruction and converts an underconstrained joint estimation problem into a stable optimization process. Specifically,

we reformulate in-hand object pose estimation as a finite template-constrained search problem, deploy an efficient neural implicit reconstruction framework for dynamic HOI sequences, and introduce a 3D relative offset prior to resolve global spatial ambiguity. This formulation enables stable pose recovery under occlusion, coherent long-sequence reconstruction, and physically consistent hand-object interactions, while achieving substantial efficiency improvements over prior methods. Extensive experiments demonstrate that AnchorHOI achieves state-of-the-art HOI reconstruction performance, delivers substantial computational speedup, and significantly improves the robustness and spatial consistency of reconstructed hand-object interactions.

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